

Supporting information for: Stark sexual dimorphism in flowering responses to vernalization in a monoecious shrub

*Methods:*

I obtained the data for this study from a 2017 experiment in which the authors observed flowering and leafout phenology on twig cuttings for 12 woody plants species exposed to 8-way fully-factorial treatment combinations of vernalization (chilling), forcing and photoperiod in growth chamber<sup>1</sup>. Detailed methodology related to the study are available at Buonaiuto and Wolkovich (2021)<sup>1</sup>. For dioecious and monoecious species included in the original study, the authors recorded both male and female flowering. Of these species, only one, *Comptonia peregrina*, opened enough male and female catkins to allow for subsequent analyses, which is the focus of this current study.

This analysis includes 96 dormant twig cuttings of *C. peregrina* collected from eight different wild stands at Harvard Forest, Petersham Massachusetts USA (42.5314°N, 72.1900°W). Twigs were collected on 25 October 2017, and transported back to the Arnold Arboretum (Boston, MA) where they were re-cut in water to minimize embolism and placed in 500 ml Erlenmeyer flasks with distilled water. Each cutting contained the buds of male catkins that were visible at the time of collection, and buds suspected of containing female catkins (n=192), though this could not be verified visually until after anthesis. The twigs were randomly assigned to a fully crossed set of eight experimental treatments; two levels of vernalization (30 versus 60 days at 4°C), two levels of forcing temperature (24°C:18°C (day/night) versus 18°C:12°C (day/night)) and two levels of photoperiod (12 versus 8 hr). The twigs were re-cut every 7–10 days and

rotated between growth chambers every 2 weeks to minimize chamber effects <sup>1</sup>. Twigs were observed every 2-3 days for three months, and male and female flowering was recorded.

I assessed the interactive effects of the three climate predictors (vernalization, forcing, and photoperiod) on the likelihood of flowering using Bayesian hierarchical models with a Bernoulli likelihood function. Because cutting from individual stands were replicated across treatment combinations, I included stand of origin as a random effect.

Because of there was complete separation in the likelihood of flowering for male catkins in response to vernalization (no flowering in low vernalization and complete flowering in high vernalization treatment), I used weakly-informative priors for the slopes and intercept parameters [normal(0,10)] to achieve model convergence.

I ran the model in R version 4.1.2 <sup>2</sup> using the packages brms<sup>3</sup> on 4 chains with for 5000 iterations each, with 4000 iteration burn-in resulting in a total of 4,000 sampling interactions across the chains. I assessed model convergence with rhats between 1.0 and 1.01, high effective sample sizes (>1500), and no divergent transitions.

## References

1. Buonaiuto, D.M., and Wolkovich, E.M. (2021). Differences between flower and leaf phenological responses to environmental variation drive shifts in spring phenological sequences of temperate woody plants. *J. Ecol.* 109, 2922–2933. <https://doi.org/10.1111/1365-2745.13708>.
2. Team, R.C. (2019). R: A Language and Environment for Statistical Computing (R Foundation for Statistical Computing).
3. Bürkner, P.-C. (2018). Advanced Bayesian Multilevel Modeling with the R Package brms. *R J.* 10, 395–411. <https://doi.org/10.32614/RJ-2018-017>.